

Name: Chinta Lakshmi Nivas Jaswanth

Module 4

1)Setting Up Junit

Java class:

```
public class Calculator {  
    public int add(int a, int b) {  
        return a + b;  
    }  
}
```

Test Class:

```
import org.junit.Test;  
import static org.junit.Assert.assertEquals;  
public class CalculatorTest {  
    @Test  
    public void testAdd() {  
        Calculator calculator = new Calculator();  
        int result = calculator.add(5, 3);  
        assertEquals(8, result);  
    }  
}
```

```
}
```

2) Assertions in Junit

```
import org.junit.Test;
import static org.junit.Assert.*;
public class AssertionsTest {
    @Test
    public void testAssertions() {
        assertEquals(5, 2 + 3);
        assertTrue(5 > 3);
        assertFalse(5 < 3);
        assertNull(null);
        assertNotNull(new Object());
    }
}
```

3) Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in Junit

Example Class:

```
public class Calculator {
    public int add(int a, int b) {
        return a + b;
    }
}
```

```
}
```

Complete Program:

```
import org.junit.Before;
import org.junit.After;
import org.junit.Test;
import static org.junit.Assert.*;
public class CalculatorTest {
    private Calculator calculator;
    @Before
    public void setUp() {
        System.out.println("Setting up...");
        calculator = new Calculator();
    }
    @After
    public void tearDown() {
        System.out.println("Cleaning up...");
        calculator = null;
    }
    @Test
    public void testAdd() {
        // Arrange
        int a = 2;
        int b = 3;
```

```
// Act
int result = calculator.add(a, b);

// Assert
assertEquals(5, result);
}
}
```

4) Mocking and Stubbing

External API:

```
public class ExternalApi {
    public String getData() {
        return "Real Data";
    }
}
```

My Service:

```
public class MyService {
    private ExternalApi api;

    public MyService(ExternalApi api) {
        this.api = api;
    }

    public String fetchData() {
        return api.getData();
    }
}
```

```
}  
}
```

Test Class:

```
import static org.mockito.Mockito.*;  
import static org.junit.jupiter.api.Assertions.*;  
import org.junit.jupiter.api.Test;  
import org.mockito.Mockito;  
public class MyServiceTest {  
    @Test  
    public void testExternalApi() {  
        ExternalApi mockApi =  
            Mockito.mock(ExternalApi.class);  
        when(mockApi.getData())  
            .thenReturn("Mock Data");  
        MyService service =  
            new MyService(mockApi);  
        String result =  
            service.fetchData();  
        assertEquals("Mock Data", result);  
    }  
}
```

5) Verifying Interactions

```
import static org.mockito.Mockito.*;
import org.junit.jupiter.api.Test;
import org.mockito.Mockito;

public class MyServiceTest {

    @Test
    public void testVerifyInteraction() {
        ExternalApi mockApi =
            Mockito.mock(ExternalApi.class);

        MyService service =
            new MyService(mockApi);

        service.fetchData();
        verify(mockApi).getData();
    }
}
```

6) Logging Error Messages and Warning Levels

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class LoggingExample {

    private static final Logger logger =
        LoggerFactory.getLogger(
            LoggingExample.class);
}
```

```
public static void main(String[] args) {  
    logger.error(  
        "This is an error message");  
    logger.warn(  
        "This is a warning message");  
}  
}
```